Data Management in R

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```
print("the girl is going to school")
```

```
## [1] "the girl is going to school"
```

sjsjhf

Learning how to execute several commands in one line

```
x <- 7; y <- 8 ; z <- x +y; # print(z)hjjj
```

We are going to learn how to comment in r

```
#subtracting 12 from 25
25-12
```

```
## [1] 13
```

Finding help in r

```
#help(print)
```

working with directories in r

```
#which directory am i working in?
getwd()
```

```
## [1] "E:/Documents/GitHub/biostatistics-portfolio/practice/r/scripts"
```

```
#setting working directory

# (working-directory line removed; open the project folder or set your own path)
```

now confirm directory

```
getwd()
```

```
## [1] "E:/Documents/GitHub/biostatistics-portfolio/practice/r/scripts"
```

Mathematical operations

```r
#sequence of 1-100
1:100
```

```
##   [1]   1   2   3   4   5   6   7   8   9  10  11  12  13  14  15  16  17  18
##  [19]  19  20  21  22  23  24  25  26  27  28  29  30  31  32  33  34  35  36
##  [37]  37  38  39  40  41  42  43  44  45  46  47  48  49  50  51  52  53  54
##  [55]  55  56  57  58  59  60  61  62  63  64  65  66  67  68  69  70  71  72
##  [73]  73  74  75  76  77  78  79  80  81  82  83  84  85  86  87  88  89  90
##  [91]  91  92  93  94  95  96  97  98  99 100
```

OR

```r
seq(1,100)
```

```
##   [1]   1   2   3   4   5   6   7   8   9  10  11  12  13  14  15  16  17  18
##  [19]  19  20  21  22  23  24  25  26  27  28  29  30  31  32  33  34  35  36
##  [37]  37  38  39  40  41  42  43  44  45  46  47  48  49  50  51  52  53  54
##  [55]  55  56  57  58  59  60  61  62  63  64  65  66  67  68  69  70  71  72
##  [73]  73  74  75  76  77  78  79  80  81  82  83  84  85  86  87  88  89  90
##  [91]  91  92  93  94  95  96  97  98  99 100
```

Modulus

```r
#gives us the remainder
5%%2
```

```
## [1] 1
```

Logical operations(return true or false; boolians)

```r
2==5 #2 is equal to 5?
```

```
## [1] FALSE
```

```r
5>2 # 5 is >2?
```

```
## [1] TRUE
```

```r
!2>5  #is 2 not equal to 5?
```

```
## [1] TRUE
```

```r
(2>1) & (100!=50)  #it only returns a true if the logical operations are correct
```

```
## [1] TRUE
```

```r
(2>1) | (100!=50)
```

```
## [1] TRUE
```

to determine the class /data type; we use the class( ) function

```r
x<-5
class("x")
```

```
## [1] "character"
```

#change variable x to a factor data type

```r
x <- 11
x
```

```
## [1] 11
```

```r
class(x)
```

```
## [1] "numeric"
```

```r
x <- as.factor(x)

class(x)
```

```
## [1] "factor"
```

```r
x
```

```
## [1] 11
## Levels: 11
```

```r
x <- as.character(x)
class(x)
```

```
## [1] "character"
```

forcing a character into numeric returns an NA

Vectors simplest data structure; can carry multiple data types we use the function c()

```r
diseases <- c("malaria","hiv","tb","htn")
diseases
```

```
## [1] "malaria" "hiv"     "tb"      "htn"
```

Matrices; 2 dimensional structures as a data frame only carries one data type

```r
matrix(1:6, nrow = 2, byrow = FALSE)
```

```
##      [,1] [,2] [,3]
## [1,]    1    3    5
## [2,]    2    4    6
```

byrow true

```r
matrix(1:6, nrow = 2, byrow = TRUE)
```

```
##      [,1] [,2] [,3]
## [1,]    1    2    3
## [2,]    4    5    6
```

```r
data("mtcars") # calling on the mtcars dataset

mtcars <- data.frame(mtcars)
```

2ND CLASS

-Loading data -install packages -exploring datasets -descriptive statistics -working with multiple datasets -renaming variables and adding labels

Loading data

```r
#using read.csv
#first get working directory
getwd()
```

```
## [1] "E:/Documents/GitHub/biostatistics-portfolio/practice/r/scripts"
```

```r
# (working-directory line removed; open the project folder or set your own path)
getwd()
```

```
## [1] "E:/Documents/GitHub/biostatistics-portfolio/practice/r/scripts"
```

ASSIGNMENT 1

##QN 1; Loading data sets

-Malaria cases, suspected, number.csv

-MALARIA_TOTAL_CASES.dta

-ITN Access.xlsx (Read both sheets)

```r
#loading( Malaria cases, suspected, number.csv using read.csv)
malaria_cases_df <- read.csv(here::here("data", "raw", "Malaria cases, suspected, number.csv"))
```

loading MALARIA_TOTAL_CASES.dta

```r
#loading MALARIA_TOTAL_CASES.dta using haven package
#haven package installed in console
library(haven)
malaria_total_cases_df <- read_dta(here::here("data", "raw", "MALARIA_TOTAL_CASES.dta"))
```

Loading ITN Access; both sheets

```r
#Loading ITN Access using the readxl package
#readxl package installed in console

library(readxl)
itn_access1 <- read_excel(here::here("data", "raw", "ITN Access.xlsx"), sheet = 1)   # read sheet 1

itn_access2 <- read_excel(here::here("data", "raw", "ITN Access.xlsx"), sheet = 2)  #read sheet 2
```

QN 2; EXPLORING THE DATASETS

```r
View(malaria_cases_df)  #view malaria_cases_df

head(malaria_total_cases_df, 10) #obtain first 10 rows of dataframe
```

```
## # A tibble: 10 x 25
##         id indicatorcode       spatialdimension spatialdimensionvaluecode
##      <dbl> <chr>               <chr>            <chr>
##  1 6653181 MALARIA_TOTAL_CASES COUNTRY          PAK
##  2 5140438 MALARIA_TOTAL_CASES COUNTRY          MWI
##  3 3504290 MALARIA_TOTAL_CASES COUNTRY          MDG
##  4 8663118 MALARIA_TOTAL_CASES COUNTRY          CAF
##  5 4520145 MALARIA_TOTAL_CASES COUNTRY          AFG
##  6 6939956 MALARIA_TOTAL_CASES COUNTRY          MLI
##  7 3804728 MALARIA_TOTAL_CASES COUNTRY          AFG
##  8 9255117 MALARIA_TOTAL_CASES COUNTRY          MOZ
##  9 2516740 MALARIA_TOTAL_CASES COUNTRY          ZAF
## 10 2556863 MALARIA_TOTAL_CASES COUNTRY          BRA
## # i 21 more variables: parentlocationcode <chr>, parentlocation <chr>,
## #   timedimension <chr>, timedim <dbl>, disaggregatingdimension1 <dbl>,
## #   disaggregatingdimension1valuecod <dbl>, disaggregatingdimension2 <dbl>,
## #   disaggregatingdimension2valuecod <dbl>, disaggregatingdimension3 <dbl>,
## #   disaggregatingdimension3valuecod <dbl>, datasourcedimension <dbl>,
## #   datasourcedimensionvaluecode <dbl>, value <chr>, numericvalue <dbl>,
## #   low <dbl>, high <dbl>, comments <dbl>, date <chr>, ...
```

```r
tail(itn_access1, 9) # obtain last 9 rows of dataframe
```

```
## # A tibble: 9 x 14
##   IndicatorCode        Indicator     ValueType ParentLocationCode ParentLocation
##   <chr>                <chr>         <chr>     <chr>              <chr>
## 1 MALARIA_ITN_COVERAGE Population w~ numeric   AFR                Africa
## 2 MALARIA_ITN_COVERAGE Population w~ numeric   AFR                Africa
## 3 MALARIA_ITN_COVERAGE Population w~ numeric   AFR                Africa
```

```
## 4 MALARIA_ITN_COVERAGE Population w~ numeric   AFR                Africa
## 5 MALARIA_ITN_COVERAGE Population w~ numeric   AFR                Africa
## 6 MALARIA_ITN_COVERAGE Population w~ numeric   AFR                Africa
## 7 MALARIA_ITN_COVERAGE Population w~ numeric   AFR                Africa
## 8 MALARIA_ITN_COVERAGE Population w~ numeric   AFR                Africa
## 9 MALARIA_ITN_COVERAGE Population w~ numeric   AFR                Africa
## # i 9 more variables: 'Location type' <chr>, SpatialDimValueCode <chr>,
## #   'Period type' <chr>, Period <dbl>, IsLatestYear <lgl>,
## #   FactValueNumeric <dbl>, Value <dbl>, Language <chr>, DateModified <chr>
```

QN 3; View the first 10 columns for the first 30 countries in the MALARIA_TOTAL_CASES. dataset

```
#using indexing
malaria_total_cases_df[1:30,1:10]
```

```
## # A tibble: 30 x 10
##         id indicatorcode       spatialdimension spatialdimensionvaluecode
##      <dbl> <chr>               <chr>            <chr>
##  1 6653181 MALARIA_TOTAL_CASES COUNTRY          PAK
##  2 5140438 MALARIA_TOTAL_CASES COUNTRY          MWI
##  3 3504290 MALARIA_TOTAL_CASES COUNTRY          MDG
##  4 8663118 MALARIA_TOTAL_CASES COUNTRY          CAF
##  5 4520145 MALARIA_TOTAL_CASES COUNTRY          AFG
##  6 6939956 MALARIA_TOTAL_CASES COUNTRY          MLI
##  7 3804728 MALARIA_TOTAL_CASES COUNTRY          AFG
##  8 9255117 MALARIA_TOTAL_CASES COUNTRY          MOZ
##  9 2516740 MALARIA_TOTAL_CASES COUNTRY          ZAF
## 10 2556863 MALARIA_TOTAL_CASES COUNTRY          BRA
## # i 20 more rows
## # i 6 more variables: parentlocationcode <chr>, parentlocation <chr>,
## #   timedimension <chr>, timedim <dbl>, disaggregatingdimension1 <dbl>,
## #   disaggregatingdimension1valuecod <dbl>
```

QN 4; From the "ITN access" sheet of ITN Access.xlsx, display all rows for the following columns: Indicator, ParentLocation, SpatialDimValueCode, Period, and FactValueNumeric

```
#using indexing
itn_access1[1:360, c("Indicator", "ParentLocation", "SpatialDimValueCode", "Period", "FactValueNumeric"]
```

```
## # A tibble: 360 x 5
##    Indicator          ParentLocation SpatialDimValueCode Period FactValueNumeric
##    <chr>              <chr>          <chr>                <dbl>            <dbl>
##  1 Population with a~ Africa         GAB                   2023             13.0
##  2 Population with a~ Eastern Medit~ DJI                   2023             24.9
##  3 Population with a~ Africa         AGO                   2023             27.0
##  4 Population with a~ Eastern Medit~ SOM                   2023             28.5
##  5 Population with a~ Africa         ETH                   2023             28.5
```

```
##  6 Population with a~ Africa         GNQ                 2023            29.0
##  7 Population with a~ Africa         GNB                 2023            33.3
##  8 Population with a~ Africa         MRT                 2023            38.7
##  9 Population with a~ Africa         ERI                 2023            41.7
## 10 Population with a~ Africa         MWI                 2023            47.1
## # i 350 more rows
```

```
#using dplyr
itn_access1  %>% select("Indicator", "ParentLocation", "SpatialDimValueCode", "Period", "FactValueNumeri
```

QN 5; Using the str() function, determine the data type for each of the following columns: Indicator, ParentLocation, SpatialDimValueCode, Period, and FactValueNumeric in the "ITN access" sheet of ITN Access.xlsx.

```
#using indexing
str(itn_access1[c("Indicator", "ParentLocation", "SpatialDimValueCode", "Period", "FactValueNumeric")])
```

```
## tibble [360 x 5] (S3: tbl_df/tbl/data.frame)
##  $ Indicator          : chr [1:360] "Population with access to an insecticide-treated bed net (ITN) i
##  $ ParentLocation     : chr [1:360] "Africa" "Eastern Mediterranean" "Africa" "Eastern Mediterranean"
##  $ SpatialDimValueCode: chr [1:360] "GAB" "DJI" "AGO" "SOM" ...
##  $ Period             : num [1:360] 2023 2023 2023 2023 2023 ...
##  $ FactValueNumeric   : num [1:360] 13 24.9 26.9 28.5 28.5 ...
```

```
#using dplyr
library(dplyr)
str(select(itn_access1, "Indicator", "ParentLocation", "SpatialDimValueCode", "Period", "FactValueNumeri
```

```
## tibble [360 x 5] (S3: tbl_df/tbl/data.frame)
##  $ Indicator          : chr [1:360] "Population with access to an insecticide-treated bed net (ITN) i
##  $ ParentLocation     : chr [1:360] "Africa" "Eastern Mediterranean" "Africa" "Eastern Mediterranean"
##  $ SpatialDimValueCode: chr [1:360] "GAB" "DJI" "AGO" "SOM" ...
##  $ Period             : num [1:360] 2023 2023 2023 2023 2023 ...
##  $ FactValueNumeric   : num [1:360] 13 24.9 26.9 28.5 28.5 ...
```

After using the above methods; the structure of the datasets is as follows; -Indicator is character -ParentLocation is character -SpatialDimValueCode is character -Period is numeric -FactValueNumeric is numeric

QN 6; Using the summary() function, determine the mean, minimum, and maximum number of malaria cases from the Malaria cases, suspected, number.csv dataset.

```
summary(malaria_cases_df) #summarises the Malaria cases, suspected, number.csv dataset.
```

```
##  IndicatorCode       Indicator          ValueType         ParentLocationCode
##  Length:859         Length:859         Length:859         Length:859
```

```
##  Class :character   Class :character   Class :character   Class :character
##  Mode  :character   Mode  :character   Mode  :character   Mode  :character
##
##
##
##  ParentLocation     Location.type      SpatialDimValueCode   Location
##  Length:859         Length:859         Length:859          Length:859
##  Class :character   Class :character   Class :character    Class :character
##  Mode  :character   Mode  :character   Mode  :character    Mode  :character
##
##
##
##  Period.type            Period     IsLatestYear    FactValueNumeric
##  Length:859         Min.   :2015   Mode :logical   Min.   :      121
##  Class :character   1st Qu.:2017   FALSE:758       1st Qu.:   147432
##  Mode  :character   Median :2019   TRUE :101       Median :   651385
##                     Mean   :2019                   Mean   :  4877290
##                     3rd Qu.:2021                   3rd Qu.:  3413658
##                     Max.   :2023                   Max.   :164949610
##     Value              Language         DateModified
##  Length:859         Length:859         Length:859
##  Class :character   Class :character   Class :character
##  Mode  :character   Mode  :character   Mode  :character
##
##
##
```

```
summary(malaria_cases_df$FactValueNumeric) #focuses on the FactValueNumeric column
```

```
##      Min.   1st Qu.    Median      Mean   3rd Qu.      Max.
##       121    147432    651385   4877290   3413658 164949610
```

```
#MALARIA CASES
#mean was 4877290
#minimum was  121
#maximum was 164949610
```

QN 7; Using the rename function from dplyr, rename factvaluenumeric from the malaria cases from the Malaria cases, suspected, number.csv dataset to "Number of malaria cases".

```
malaria_cases_df <- malaria_cases_df %>% rename("Number of malaria cases" = FactValueNumeric)
head(malaria_cases_df)
```

```
##      IndicatorCode                        Indicator ValueType
## 1 MALARIA_SUSPECTS Number of suspected malaria cases   numeric
## 2 MALARIA_SUSPECTS Number of suspected malaria cases   numeric
## 3 MALARIA_SUSPECTS Number of suspected malaria cases   numeric
## 4 MALARIA_SUSPECTS Number of suspected malaria cases   numeric
## 5 MALARIA_SUSPECTS Number of suspected malaria cases   numeric
```

```
## 6 MALARIA_SUSPECTS Number of suspected malaria cases   numeric
##   ParentLocationCode        ParentLocation Location.type SpatialDimValueCode
## 1                EMR Eastern Mediterranean       Country                 AFG
## 2                WPR       Western Pacific       Country                 KHM
## 3                AFR                Africa       Country                 COG
## 4                AFR                Africa       Country                 ZWE
## 5                WPR       Western Pacific       Country                 VNM
## 6                AMR              Americas       Country                 BRA
##      Location Period.type Period IsLatestYear Number of malaria cases     Value
## 1 Afghanistan        Year   2023         TRUE                 1003373 1 003 373
## 2    Cambodia        Year   2023         TRUE                 1151739 1 151 739
## 3       Congo        Year   2023         TRUE                 1155691 1 155 691
## 4    Zimbabwe        Year   2023         TRUE                 1197321 1 197 321
## 5    Viet Nam        Year   2023         TRUE                 1248412 1 248 412
## 6      Brazil        Year   2023         TRUE                 1407046 1 407 046
##   Language             DateModified
## 1       EN 2024-12-10T05:00:00.000Z
## 2       EN 2024-12-10T05:00:00.000Z
## 3       EN 2024-12-10T05:00:00.000Z
## 4       EN 2024-12-10T05:00:00.000Z
## 5       EN 2024-12-10T05:00:00.000Z
## 6       EN 2024-12-10T05:00:00.000Z
```